Decorated points and tetrahedral geometries

Human Compression project

Abstract

The two large parabolics in the tetrahedral construction are stabilizers of nonzero vectors modulo a subgroup of scalars. Their incidence is therefore a determinant condition on decorated projective points. This describes the relevant two types uniformly and exposes the dependence on a choice of coordinates in each fibre. The same affine coordinates give the full period lattices of the two toroidal residues. We distinguish those lattices from the normal closure of a single relation, which need not have the correct index. Two orbits of decorated tetrahedra give all four types, their flags and their correlations. The small characteristic is treated separately. The construction recovers the two families of hypertopes while correcting their printed geometric coordinates and some of their period labels.

1 The parabolic groups

Let $p \neq 3$ be prime. Let $F = \mathbb{F}_p(e)$, where $e^2 + e + 1 = 0$ and $e \neq 1$, and put $q = |F|$. Thus $q = p$ when $p \equiv 1 \pmod 3$, and $q = p^2$ when $p \equiv 2 \pmod 3$. Write

$$U(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad V(t) = \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}, \qquad E = \begin{pmatrix} e & 0 \\ 0 & e^2 \end{pmatrix}.$$

All matrix groups below are read in $G = \mathrm{PSL}_2(F)$; the displayed matrices have determinant one. The source marking is

$$a_1 = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \quad a_2 = \begin{pmatrix} e & e^2 \\ 0 & e^2 \end{pmatrix}, \quad a_3 = \begin{pmatrix} e^2 & e \\ 0 & e \end{pmatrix}.$$

For its parabolics $G_1 = \langle a_2, a_3 \rangle$ and $G_0 = \langle a_1^{-1}a_2, a_1^{-1}a_3 \rangle$, direct multiplication gives

$$a_2 = U(1)E, \quad a_3 = U(1)E^{-1}, \qquad a_1^{-1}a_2 = V(1)E, \quad a_1^{-1}a_3 = V(1)E^{-1}.$$

In particular the lower-left entry of $a_1^{-1}a_3$ is e^2; the isolated formula on source page 8 printing e in that position is a typographical error.

Lemma 1. *The two marked parabolics are*

$$G_1 = U(F) \rtimes \langle E \rangle, \qquad G_0 = V(F) \rtimes \langle E \rangle.$$

They have order $3q$*, and* $G_0 \cap G_1 = \langle E \rangle$*. The three source generators generate* G*.*

Proof. In either pair $X = T(1)E$, $Y = T(1)E^{-1}$, one has $X^{-1}Y = E$ and $XE^{-1} = T(1)$. Conjugation by E multiplies the parameter of $U(t)$ by e^2 and that of $V(t)$ by e. The additive spans of $1, e$ and of $1, e^2$ are both F. Thus all translations occur, and no further elements occur since the stated affine groups contain the generators. Their complements have order three and meet their translation groups trivially. An upper and a lower triangular matrix of the displayed forms can agree projectively only when both off-diagonal entries vanish; this proves the intersection statement. Finally the generated group contains every $U(t)$ and

$V(t)$. Elementary row operations express every determinant-one matrix as a product of these. If a leading pivot vanishes, the matrix $U(1)V(-1)U(1)$ exchanges the two coordinate lines; otherwise one eliminates the lower entry and then the upper entry. Diagonal matrices are products as well, since

$$U(t)V(-t^{-1})U(t) = \begin{pmatrix} 0 & t \\ -t^{-1} & 0 \end{pmatrix} \quad (t \neq 0),$$

and multiplying two such matrices gives any determinant-one diagonal matrix. This proves generation without a subgroup classification. □

2 Decorations and incidence

Put $d = \gcd(2, q-1)$ and $K = \{\pm 1, \pm e, \pm e^2\} \leq F^\times$, so $|K| = 3d$. A decorated point is an equivalence class

$$[v]_K = Kv, \qquad v \in F^2 \setminus \{0\}.$$

Forgetting the decoration gives its projective line Fv. Each fibre has $(q-1)/(3d)$ elements and is a torsor for $F^\times/K$: scalar multiplication acts freely and transitively. There is no preferred element in a fibre.

Theorem 2. *The source cosets of types* 0 *and* 1 *are two copies of the decorated points. Under the identifications*

$$G_0 g \longmapsto [g^{-1}(0,1)^T]_K, \qquad G_1 g \longmapsto [g^{-1}(1,0)^T]_K,$$

their incidence is

$$\det(v, w) \in K. \tag{1}$$

It is independent of the vector representatives. Each object has q neighbours, and this bipartite incidence graph is connected.

Proof. The stabilizer in $\mathrm{SL}_2(F)$ of $[(1,0)^T]_K$ consists of the upper triangular matrices $\left(\begin{smallmatrix} k & b \\ 0 & k^{-1} \end{smallmatrix}\right)$ with $k \in K$ and $b \in F$. Its projective image is exactly G_1. The analogous lower triangular stabilizer is G_0. The action on decorated points is transitive: a nonzero vector can be completed to a matrix of determinant one. The orbit–coset identifications follow, with the inverse accounting for the source's right cosets.

If two cosets intersect, a common representative takes their two base vectors to representatives of $[v]_K, [w]_K$. Their determinant is then 1 or -1 before multiplying by allowed scalar factors, so (1) holds. Conversely, if $\det(v, w) \in K$, one can multiply one vector by an element of K to make the determinant one. The resulting ordered pair is a matrix in $\mathrm{SL}_2(F)$ and supplies a common coset representative. Multiplying either vector by a member of K cannot change condition (1).

Fix $v = (1,0)^T$. Every adjacent class has a unique representative of the form $(a,1)^T$, with $a \in F$. This proves the degree. To prove connectedness, first take representatives v, v' on distinct projective lines in the same copy. The two independent linear equations

$$\det(v, w) = 1, \qquad \det(v', w) = 1$$

have a solution $w \neq 0$, which gives a path of length two between their decorated classes. If two classes lie on the same projective line, choose a third class on a different line and apply this argument twice. All vertices in one copy are connected. Every vertex in the other copy has a neighbour, so the whole graph is connected. □

Choose a vector b_z on each projective line z. Writing its decorated points as $[tb_z]_K$, with $tK \in F^\times/K$, gives the explicit coordinate formula

$$[tb_z]_K * [ub_w]_K \quad \Longleftrightarrow \quad tu \det(b_z, b_w) \in K. \tag{2}$$

Changing the chosen vector b_z changes the scalar coordinate at z and the determinant factors together. This is the dependence on the choice of representatives that a description by a product set must retain.

Section 6, equation (4) of the fixed source instead makes two objects (t, z) and (u, w) incident exactly when $t = u$ and $z \neq w$. For $m = (q-1)/(3d) > 1$, this gives m disjoint connected components, each a complete bipartite graph on two copies of the projective line with the matching deleted. The actual coset truncation is connected by Theorem 2; hence those two incidence rules cannot be isomorphic. For example $q = 13$ gives two components under the printed rule and one under (1). The correction concerns the geometric description in Section 6; it does not refute the separate coset construction of Section 4.

3 Two orbits of decorated tetrahedra

Write $x = [(1,0)^T]_K$, $y = [(0,1)^T]_K$ and $z = [(1,1)^T]_K$. Put

$$T = \{x, y, z\}, \qquad d_2 = [(-e^2, 1)^T]_K, \quad d_3 = [(-e, 1)^T]_K, \qquad D_i = T \cup \{d_i\}.$$

The four points of either D_i have different projective lines. Every determinant of two distinct representatives in D_i belongs to K. For example the determinants involving $(1,1)^T$ and $(-e^2,1)^T$ give $1 + e^2 = -e$; the other five determinants are ± 1 or $\pm e^2$. The calculation for D_3 exchanges e, e^2. Let $\mathcal{D}_i$ be the G-orbit of D_i.

Lemma 3. *The stabilizers of D_2, D_3 are respectively $G_2 = \langle a_1, a_3 \rangle$ and $G_3 = \langle a_1, a_2 \rangle$, each acting as A_4 on its four decorated points. Suppose p is odd. Then the stabilizer of T is $\langle a_1 \rangle \cong C_3$, the two tetrahedron orbits are disjoint, and each triple in a tetrahedron of one orbit belongs to exactly one tetrahedron of the other orbit.*

Proof. On either tetrahedron a_1 cycles x, y, z and fixes its fourth point. On D_2, a_3 fixes x and cycles y, z, d_2; on D_3, a_2 fixes x and cycles y, z, d_3. These assertions follow by multiplying the four column vectors; all scalar factors are in K. The two displayed three-cycles generate A_4. A projective linear transformation fixing three different projective lines is the identity, so these actions are faithful and both subgroups have order twelve.

Consider an element stabilizing T. If it fixes each of x, y, z, a lift to $\mathrm{SL}_2(F)$ is diagonal, and fixing z forces its diagonal entries to agree. It is scalar and hence the identity in G. Thus the stabilizer embeds in S_3 and contains the three-cycle a_1. An element transposing x, y and fixing z would have a lift

$$\begin{pmatrix} 0 & k \\ k & 0 \end{pmatrix}, \qquad k \in K,$$

with determinant $-k^2 = 1$. For odd p, the square subgroup of K has order three and cannot contain -1. There is no such transposition; composing any odd permutation with a power of a_1 reduces to this case. The stabilizer is therefore C_3.

For p odd, an element stabilizing D_i can first be composed with an element of the displayed A_4 to fix d_i. It then stabilizes T and already belongs to that A_4. This proves the full stabilizer assertion. If a translate gD_i contains T, transitivity of G_i on the four triples of D_i allows us to choose $h \in G_i$ with $ghT = T$. Consequently $gD_i = ghD_i = D_i$. Thus D_i is the unique member of its orbit containing T. In particular D_2, D_3 are in different orbits. Transporting this argument by G proves the assertion for every triple in either orbit.

The remaining characteristic is $p = 2$, so $q = 4$ and $K = F^\times$. The decorated points are the five ordinary projective points. The group G has order sixty and acts faithfully on them, with the three generators acting by even permutations. It is A_5. Both D_i are four-subsets, and their orbits are the five four-subsets. Their stabilizers have order twelve and are precisely the already exhibited G_i. The assertions specific to odd p are not used in this case. □

Define a four-type incidence system as follows. Types $0, 1$ are two copies of the decorated points; types $2, 3$ are the orbits $\mathcal{D}_2, \mathcal{D}_3$, retaining the type even if the orbits coincide. Between different types use

types	incidence
$0, 1$	$\det(v, w) \in K$,
i, j $(i < 2 \le j)$	$[v]_K \in D$,
$2, 3$	$\lvert D \cap D' \rvert = 3$.

(3)

Theorem 4. *The incidence system (3) is the source's right-coset system $\Gamma(G; (G_i)_{i=0}^3)$. It is a hypertope with tetrahedral diagram having all six bonds labelled three. The group G has two free orbits on its chambers; adjacent chambers belong to different orbits. All rank-three flags form one free G-orbit for each choice of three types.*

Proof. The point stabilizers and the first incidence rule were proved in Theorem 2. Lemma 3 identifies the other two stabilizers. Thus the object map sends a right coset $G_i g$ to g^{-1} applied to the corresponding base object y, x, D_2, D_3. Every incident point–tetrahedron pair is in the orbit of the corresponding base pair, because A_4 is transitive on the four vertices. For odd p, every tetrahedron of type 2 has exactly four neighbours of type 3, one for each triple. Its A_4 stabilizer is transitive on these triples. Hence these pairs too form exactly the orbit of the base pair (D_2, D_3). Pair incidence in a coset system is exactly the orbit of its base pair, proving the claim about all six relations.

For the moment suppose p is odd. A flag of types 012 or 013 is a tetrahedron with two ordered distinct vertices. Its stabilizer A_4 acts regularly on the twelve ordered pairs of distinct vertices. These rank-three flags therefore form one free orbit. A flag of types 023 or 123 consists of adjacent tetrahedra and a point of their common triple. The stabilizer of the pair is C_3, acting regularly on that triple. These flags likewise form one free orbit.

There are exactly two extensions of each such flag to a chamber. For types 023 or 123, the missing point can be either of the other two vertices in the common triple. For types 012 or 013, the missing tetrahedron is specified by a triple of the given tetrahedron containing the two given points. There are exactly two choices, giving different opposite tetrahedra. Every incident pair extends to one of these flags: for a point and a tetrahedron choose another vertex; for adjacent tetrahedra choose a point in their common triple; and the point pair case follows by transporting the base point pair. Every object has an incident pair. Thus this is a geometry, and the two-extension calculation proves thinness.

The stabilizer of each base incident pair is cyclic of order three. For pair types 01, 02, 03, 12, 13 and 23, these groups are respectively

$$\langle E \rangle, \quad \langle a_1^{-1} a_3 \rangle, \quad \langle a_1^{-1} a_2 \rangle, \quad \langle a_3 \rangle, \quad \langle a_2 \rangle, \quad \langle a_1 \rangle.$$

For point–tetrahedron pairs this follows from the A_4 action; the other two cases were proved above. Since triple stabilizers are trivial, every rank-two residue has three objects of each of its two types. Thinness makes its graph a simple bipartite graph of degree two, hence a six-cycle. This proves both connectedness of these residues and the bond labels.

Each rank-three residue is the coset geometry inside its object stabilizer G_i with parabolics $G_i \cap G_j$. This assertion includes incidence: a pair in such a residue, together with the fixed object, is a rank-three flag, so the transitivity just proved puts it in the orbit of the base

flag under G_i. These three cyclic parabolics generate G_i in every case, as the displayed generators show. Their coset incidence graph is connected: writing a group element as a product of elements of these parabolics gives a path between the corresponding chambers and their objects. The whole geometry is connected already on types $0, 1$, and every tetrahedron has point neighbours. All required residues are therefore connected.

There are exactly $2|G|$ chambers, because each rank-three flag has two extensions and there are $|G|$ flags of any fixed three types. Chamber stabilizers are trivial. If adjacent chambers were in the same orbit, a group element taking one to the other would fix their common rank-three flag and would be the identity, a contradiction. This proves the orbit assertions.

For $q = 4$, identify each tetrahedron with its missing projective point. All four types are then copies of the same five-element set, with incidence meaning unequal labels. A chamber is an ordered choice of four different labels. Every three-label flag has two extensions; every rank-two residue is a six-cycle, and every rank-three residue is three copies of four labels with unequal-label incidence, which is connected. The A_5 action on ordered triples of different labels is free and transitive. These observations give all the same conclusions directly, including the coset identification by the stabilizers. □

The six pair stabilizers are distinct, and their intersections are trivial whenever a third base object is fixed. Together with $G_i \cap G_j$ as displayed, this proves the full intersection property for the rotation marking: a subgroup indexed by two labels is one of these cyclic groups, one indexed by three is a maximal parabolic, and the intersections are their indexed meets. For example intersecting a cyclic group with a maximal parabolic that does not contain it fixes three base objects and is trivial. Two distinct cyclic groups have trivial intersection by the same argument. The generators are independent since omitting one leaves a proper maximal parabolic. This also recovers the source's C^+-group claims, rather than assuming them to construct the geometry.

4 Residues and symmetries

The residue of a tetrahedron is three copies of its four vertices: the opposite tetrahedron is labelled by the one missing vertex of its common triple. Incidence again means unequal labels. Its rotation group is A_4; the two triangular translation words are distinct nonidentity double transpositions and generate its Klein four subgroup. Their squares are the identity, so the translation kernel contains $2\mathbb{Z}^2$. Its index is four because the image is that Klein four group. The kernel is therefore exactly $2\mathbb{Z}^2$, giving the marked hypermap $(3, 3, 3)_{(0,2)}$.

The residue of a point has rotation group G_0 or G_1, with the ordered generators used in the next section. Its actual incidence is the internal coset incidence by the rank-three flag argument in Theorem 4. The classical triangular rotation group maps onto it with a translation kernel. Therefore the residue is the triangular toroidal quotient by precisely that kernel, not merely by a relation satisfied in the finite group. In either point residue the three cyclic parabolics, in order, are $\langle X^{-1}Y \rangle$, $\langle Y \rangle$ and $\langle X \rangle$; they have order three and trivial pairwise intersections. Thus this is the standard marked C^+ coset construction. The residue stabilizer has two free chamber orbits, inherited from the two global orbits, and adjacent chambers are in opposite orbits. The full kernels and the resulting correction of the printed period parameters are calculated below.

Consider the two linear maps

$$A = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Both have determinant $-1 \in K$ and normalize $\mathrm{SL}_2(F)$. They act on decorated points and tetrahedra, preserve all the relations (3), and exchange D_2, D_3. The latter assertion follows by applying each map to their four displayed vectors. Thus A, accompanied by the type permutation (2 3), is a correlation μ. It sends the base chamber (y, x, D_2, D_3) to (z, x, D_2, D_3), its type-zero neighbour. The map B, accompanied by (0 1)(2 3), is a correlation η fixing the base chamber as a set. Both normalize the G action. Accordingly μ exchanges its two chamber orbits, whereas η preserves them. This proves the improper and proper correlation assertions by the same object model.

When $q = p^2$, Frobenius $\sigma : t \mapsto t^p$ exchanges e, e^2. On the object model it preserves determinant membership in K, fixes x, y, z and exchanges D_2, D_3. Accompanied by the type permutation (2 3) it is a correlation fixing the base chamber. Its composition with μ is type preserving and takes the base chamber to its type-zero neighbour. Together with G it acts transitively on all chambers. The hypertope is regular.

For $q = p$, the kernel calculation below shows that neither marked point residue admits an automorphism inverting its two standard rotation generators. The acquired theorem on two chamber orbits [3], applied with the preceding actual coset and chamber identifications, says that the residue is regular exactly when this inversion descends. Its proof labels all chambers $(g, +), (g, -)$ and uses the explicit reflection $(g, \pm) \mapsto (\theta(g), \mp)$ for an inverting automorphism θ; the converse follows by conjugating the rotations by a normalized chamber reflection. The non-invariant kernel therefore makes the point residue chiral. A regular geometry has regular residues: chamber transitivity restricts to chamber transitivity in the stabilizer of the fixed flag. Hence the full geometry is not regular. Its two G-orbits and thin residual connectedness show that it is chiral, with full type-preserving group G. Here we use the elementary gallery fact that in a thin residually connected geometry any type-preserving automorphism fixing a chamber is the identity: propagate across its uniquely determined adjacent chambers. Residual connectedness gives gallery connectedness by induction on rank. In particular the full group cannot split either transitive G-orbit or contain an additional element preserving both orbits.

For $q = 4$ the full correlation group is $S_5 \times S_4$. Indeed the common permutation of the five labels and any permutation of the four types preserve unequal-label incidence. Conversely, remove the induced type permutation from a correlation. For two different types, the unique non-neighbour of a label is its equal label in the other copy. The permutations on the four copies must therefore agree. The type-preserving group is S_5, and the given rotation group is $A_5 = \mathrm{PSL}_2(4)$. This corrects the final isomorphism in the source's first remark on page 19: $S_5 \times S_4$ is not $\mathrm{PSL}_2(4) \times S_4$.

5 The complete translation kernels

Use the source's ordered generators X, Y for either G_1 or G_0. In the standard triangular translation coordinates put

$$u = XY^{-1}X, \qquad v = XY.$$

The triangular rotation presentation and its translation lattice $\mathbb{Z}^2$ are retained classical inputs, already developed in the acquired hexagonal construction. The preceding matrices give

	u	v	image of $u^a v^b$
G_1	$U(1)$	$U(-e)$	$U(a - eb)$
G_0	$V(1)$	$V(-e^2)$	$V(a - e^2 b)$.

(4)

The complement still has order three, so the kernel of either marked triangular presentation is wholly in its translation subgroup.

Theorem 5. *If $q = p^2$, the period lattice of either residue group is $p\mathbb{Z}^2$. If $q = p$, choose integer representatives e_0, e_1 of $e, e^2 \in \mathbb{F}_p$. The two period lattices are respectively*

$$L_1 = \langle (p,0), (e_0,1) \rangle_{\mathbb{Z}}, \qquad L_0 = \langle (p,0), (e_1,1) \rangle_{\mathbb{Z}}. \tag{5}$$

Both have index p. In contrast, the normal translation lattice generated by the single source vector $(e_0, 1)$ has index $e_0^2 + e_0 + 1$, which can be larger than p.

Proof. When $e \notin \mathbb{F}_p$, the elements $1, e$ are linearly independent over $\mathbb{F}_p$, as are $1, e^2$. Equation (4) therefore has kernel exactly $p\mathbb{Z}^2$. When $e \in \mathbb{F}_p$, the equation $a - e_0 b \equiv 0 \pmod{p}$ is equivalent to $(a, b) = c(p, 0) + b(e_0, 1)$. This proves the first lattice in (5); the other is identical with e_1 substituted. Their determinants are p.

Conjugation by the triangular order-three complement acts on translation coordinates by

$$R(a, b) = (-a - b, a), \qquad R^2 + R + I = 0.$$

One may replace R by R^{-1} without changing an orbit lattice. The normal closure of a translation vector (a, b) is its orbit span. The two columns (a, b) and $R(a, b)$ have determinant $a^2 + ab + b^2$. Thus the orbit span of $(e_0, 1)$ has index $e_0^2 + e_0 + 1$. It is contained in the actual kernel by (4), but equality requires equality of these indices. □

For a concrete discrepancy take $p = 19$ and $e = 7$; then $7^2+7+1 = 57$. The actual parabolic group has order $3p = 57$, whereas the single-period presentation printed as $(3, 3, 3)_{(7,1)}$ has order $3 \cdot 57 = 171$. The other printed vector $(1, 7)$ has the same norm. The missing condition is the second lattice generator in (5). In the G_1 coordinates a suitable single orbit generator is $(2, 3)$: its norm is 19, and $2 - 7 \cdot 3 \equiv 0 \pmod{19}$. Its orbit span therefore equals L_1. In the G_0 coordinates $(3, 2)$ works, since $e^2 = 11$ and $3 - 11 \cdot 2 \equiv 0 \pmod{19}$.

The failure of the printed period vector does not invalidate the chirality argument. To make this precise without choosing a norm representation, an automorphism inverting both X, Y would send

$$u \longmapsto u^{-1}, \qquad v \longmapsto u^{-1} v.$$

These identities follow directly from the matrices, or from the triangular affine presentation. It would therefore preserve the kernel under $(a, b) \mapsto (-a - b, b)$. For $q = p$, this map takes $(e_0, 1) \in L_1$ to $(e_1, 1) \notin L_1$, since $e^2 \neq e$. It similarly fails to preserve L_0. Thus no such automorphism exists for either marked residue group. When $q = p^2$, the lattice $p\mathbb{Z}^2$ is preserved. These are the exact kernel statements needed for the residue symmetry test. Section 4 identified the actual residue geometry with the corresponding toroidal quotient.

6 Interfaces with the source's intermediate lemmas

The general coset construction and its transitivity on incident pairs are the acquired Tits construction. Its right action need not be faithful for arbitrary input subgroups; faithfulness here follows from the trivial chamber stabilizer. Source Proposition 2 also has a short general proof. With at least two rotation generators, every generator belongs to one of the maximal parabolics, so their union generates the group. Chambers obtained by multiplying successive such factors share an object. Every object belongs to a chamber, giving a connected incidence graph.

The coset transversals in source Lemma 9 can also be read from the object model. For $j = 0, 1$, the translation group $V(F)$ or $U(F)$ is a transversal of each cyclic subgroup $G_i \cap G_j$ in G_j: it has order q, the intersection with a subgroup of order three is trivial, and $|G_j| = 3q$. For $j = 2, 3$, the displayed three-cycles on D_j show that the source's four listed representatives send the relevant fixed vertex to its four different vertices. They are therefore transversals

too. This proves the literal neighbour lists of Lemma 9. Source Proposition 10 is now an immediate consequence of transitivity on rank-three flags: an object incident to two base objects lies in their common stabilizer orbit.

For completeness, the identity behind source Lemma 18 is

$$AV(c)Aa_1 = \begin{pmatrix} -c & 1 \\ -c-1 & 1 \end{pmatrix} = U(1)V(-c-1) = a_1V(-c).$$

Because $U(1) \in G_1$ and $a_1 \in G_2 \cap G_3$, and both $c \mapsto -c-1$ and $c \mapsto -c$ permute F, it gives exactly the three stated coset-set equalities. This calculation is no longer needed to prove that μ preserves incidence: the determinant and membership formulas prove that at once.

7 Attribution and scope

The field family and three matrices are those of Catalano, Fernandes, Hubard and Leemans [1]. The affine descriptions of the parabolics are their Lemma 6, here derived from two common identities. Vector stabilizers, determinant coordinates and the triangular translation group are standard ingredients. The addition is the common model of decorated points and tetrahedra, which derives the flag and correlation assertions from their small stabilizers. The incidence formula and complete kernel calculation also identify two discrepancies in the fixed edition. The chamber inversion theorem is acquired from the existing common theory, not reproved for this family. No historical priority is claimed for these methods. The four-type construction and its proper and improper correlations follow from the same decorated tetrahedra. The complete source disposition and independent review are recorded separately; this working manuscript does not itself certify their completion. It gives no new conjecture answer or compression measurement.

The second remark on source page 19 states that the finite groups $G^+ = \mathrm{PSL}_2(q)$ are index-four subgroups of the infinite hyperbolic Coxeter group $[6, 3, 6]$. Read literally this is impossible: a group with a finite subgroup of finite index is finite. A construction using a universal subgroup and then finite quotients would require a different statement. We do not silently substitute that possible interpretation for the printed one.

References

[1] D. Catalano, M. E. Fernandes, I. Hubard and D. Leemans, *Hypertopes with tetrahedral diagram*, Electronic Journal of Combinatorics 25(3) (2018), P3.22. Fixed published edition, catalogue L041, Lemmas 6 and 14, Section 6, equation (4).

[2] Human Compression project, *Hexagonal extensions of toroidal maps and hypermaps*, acquired triangular rotation and translation presentations.

[3] Human Compression project, *Residue stabilizers and distinguished generators*, Section 6, and *Reconstructing a reflection from two chamber orbits*, accepted common theory, topic-chamber/v2, Section 2. The latter makes explicit the criterion of Fernandes, Leemans and Ivić Weiss, *Highly symmetric hypertopes*, Aequationes Mathematicae 90 (2016), Theorem 8.2.